

2022 Ph H2 Q11

A student investigates the relationship between potential difference and current in a resistor.

(a)

State the relationship between potential difference and current in the resistor

✓ Ohm's Law:

$$V = IR$$

This states that, for a **resistor at constant temperature**, the **potential difference across it is directly proportional to the current** through it.

(b)

Explain why the student switches off the power supply between readings

✓ To prevent the resistor from heating up.

Heating the resistor would increase its **temperature**, which could **change its resistance** and **invalidate Ohm's Law** (which assumes constant temperature).

(c)

Use the graph to determine the resistance of the resistor

From the graph (not shown here but typically linear), take two points to calculate the gradient.

Assume:

- Point 1: $V = 2.0 \text{ V}$, $I = 0.40 \text{ A}$
- Point 2: $V = 4.0 \text{ V}$, $I = 0.80 \text{ A}$

$$R = \frac{V}{I} = \frac{4.0 - 2.0}{0.80 - 0.40} = \frac{2.0}{0.40} = 5.0 \Omega$$

(d)

Explain what is meant by a variable resistor

✓ A variable resistor is a component whose **resistance can be manually adjusted**.

✓ It is used to **change the current** in a circuit without needing to alter voltage.

🧠 Revision Tips

- Use $V = IR$ when the graph of V vs. I is a straight line through the origin
- The **gradient** of a V vs. I graph gives the **resistance**
- Resistors heat up as current flows; high temps may affect their resistance — hence, **switch off between readings**
- A variable resistor = **rheostat** = excellent tool for investigating how current changes with resistance